self-hosting provides greater control over your environment, it also demands a higher level of technical knowledge and ongoing maintenance efforts. You will have to take a more hands-on approach with deploying and delivering your app to users and make sure that the operating systems that host your code are configured as well.

Overall, when considering deployment options for your AI web application, it's essential to weigh the pros and cons of each approach. Each project is unique, with varying requirements based on factors such as scale, budget, technical expertise, and specific use cases. Therefore, there is no one-size-fits-all solution. Certainly, though, you need to define your priorities regarding security, compliance, and reliability. This can be achieved by establishing a production checklist.

9.6.2 Production deployment checklist

Before deploying any web application to a production environment, it's crucial to have a production checklist in place. This checklist ensures all necessary steps are accounted for and executed effectively. Scalability requirements vary significantly based on the anticipated traffic volume, and what works for a small-scale application won't suffice for a large-scale, high-traffic system. We'll start with the requirements based on expected traffic.

TRAFFIC-BASED REQUIREMENTS

Your approach to deployment should be driven by the expected load on your application. For example, let's examine three distinct tiers, each with its unique requirements for monitoring, scaling, security, and backup procedures:

- *Tier 1: Small-scale applications (<10,000 requests/day)*—For applications with a relatively low traffic volume, the focus is on establishing a reliable and secure foundation. At this tier, you should implement
  - *Basic monitoring and alerting*—Set up fundamental monitoring tools to track the application's uptime and response times. Typically, those are provided by a hosting provider like Vercel.
  - *Simple horizontal scaling*—Ensure that you can easily add more instances of your application to handle increased load.
  - *Standard security measures*—Implement essential security practices such as HTTPS, input validation, and protection against common web vulnerabilities.
  - *Daily backups with a short retention policy*—Regularly back up your application's data and configuration to prevent data loss in case of a failure.
  - *Basic error tracking*—Use error-tracking tools to identify and resolve problems promptly, ensuring a stable user experience.
- *Tier 2: Medium-scale applications (10,000–100,000 requests/day)*—As traffic increases, you'll need more sophisticated measures to ensure optimal performance and reliability. At this tier, you should enhance your infrastructure with
  - *Basic CDN implementation*—Use a CDN like Cloudflare to distribute static assets and reduce latency for users across different geographic regions. Some